Université de Haute-Alsace
Faculté des Sciences et Techniques

M2 – IM - IMDS - MIAGE

Statistical and Machine Learning

HW1 : Introduction and Learnability

Expected Deliverables

Your submission should consist of a single Jupyter notebook (`.ipynb`) containing:

- well-structured and documented code;
- appropriate visualizations;
- comments interpreting each experiment;
- a short conclusion summarizing the differences between the PAC and Agnostic PAC settings based on your experimental observations.

The completed notebook must be submitted by uploading it to the course GitHub repository. Unless otherwise specified, the submission deadline is **one week after the HW is distributed**.

Objective

The objective of this homework is to experimentally illustrate the difference between the *PAC* and the *Agnostic PAC* learning frameworks introduced during the lectures.

Throughout this homework, you will implement a simple Empirical Risk Minimization (ERM) algorithm for a hypothesis class of threshold classifiers. You will first study the realizable setting, where the data perfectly follows the target function, and then extend your implementation to the agnostic setting by introducing label noise.

Rather than following a fixed set of parameters, you are expected to design your own experiments. In particular, you should choose appropriate values for the target threshold θ, the accuracy parameter ε, the confidence parameter δ, the label noise rate η, the training sample sizes, and the number of experimental repetitions. Your conclusions should be supported by the experimental results you obtain.

No machine learning libraries (e.g., `scikit-learn`) should be used for the learning algorithm.

Learning Problem

Consider the input space

$$\mathcal{X} = [0, 1].$$

The target function is an unknown threshold classifier

$$f_\theta(x) = \begin{cases} 1, & x > \theta, \\ 0, & \text{otherwise}, \end{cases}$$

where $\theta \in (0, 1)$ is fixed.

Training examples are generated by sampling points uniformly from $\mathcal{X}$.

In the realizable setting, labels are generated according to

$$y = f_\theta(x).$$

In the agnostic setting, each label is independently flipped with probability η.

Tasks

Your notebook should contain the following sections.

1. **Data Generation**

   Generate training datasets for different sample sizes

   $$m \in \{5, 10, 20, 50, 100, 200\}.$$

   Visualize at least one generated dataset.

2. **Implementation of ERM**

   Implement an ERM algorithm over the hypothesis class of threshold classifiers.

   Explain the strategy used to determine the optimal threshold.

3. **Performance Evaluation**

   For every training set,

   - compute the empirical error;
   - compute the true error using a sufficiently large independently generated test set.

   Compare both quantities.

4. **Experimental Verification of PAC Learning**

   For every training size m,

   repeat the experiment multiple times (e.g., 500 or 1000 repetitions).

   Estimate

   $$\Pr(L(h) > \varepsilon)$$

   for a fixed value of ε.

   Study how this probability evolves as the number of training examples increases.

5. **From PAC to Agnostic PAC**

   Repeat the previous experiments after introducing label noise.

   Compare the realizable and agnostic settings.

   In particular, discuss the following questions:

- Can the empirical error still reach zero?
- Does increasing the number of training examples still improve the generalization performance?
- How do your observations relate to the PAC and Agnostic PAC learning frameworks introduced during the lectures?